Methods and Constructions for Ramsey Theory

Iyad El Khoury

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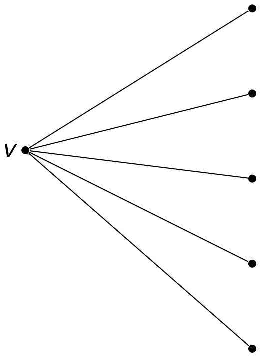
Introduction

Ramsey Numbers

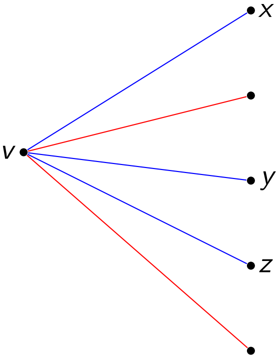
Let $n \in \mathbb{N}$ and $k_1, k_2, \ldots k_n \in \mathbb{N}$.

The Ramsey number $R(k_1, \ldots, k_n)$ is the smallest $r \in \mathbb{N}$ such that for any edge coloring using n colors of the complete graph on r vertices, there is a monochromatic clique K_{k_i} for some $i \in \{1, \ldots, n\}$.

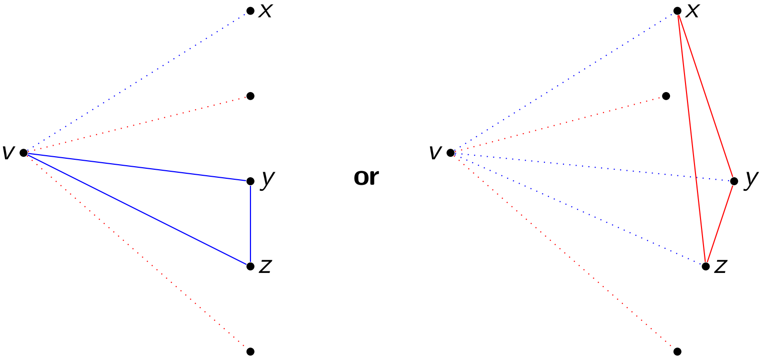
An example with $R(3,3)$



$R(3,3)$



$R(3,3)$



We have proved $R(3,3) \leq 6$
It is simple to check it is larger than 5.

Quantitative estimates

$r \backslash s$	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2		2	3	4	5	6	7	8	9
3			6	9	14	18	23	28	36
4				18	25	36–40	49–58	59–79	73–105
5					43–46	59–85	80–133	101–193	133–282
6						102–160	115–270	134–423	183–651
7							205–492	219–832	252–1368
8								282–1518	329–2662
9									565–4956

Quantitative estimates

Ramsey numbers are very hard to compute, can we give a simple equivalent?

Can we say things about the structure of extremal colorings?

Probabilistic lower bound

A first lower bound

Theorem

For any $k, n \in \mathbb{N}$, $R(k,k) > n - \frac{\binom{n}{k}}{2^{\binom{k}{2}-1}}$.

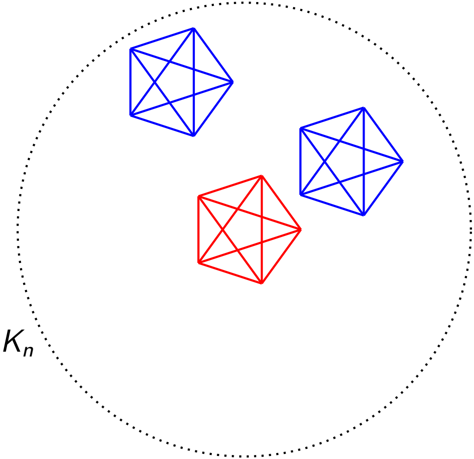
Proof

- Consider a uniform edge coloring of K_n.
- Let $X := \#\{K_k \subset K_n : K_k \text{ monochromatic}\}$.

$$\mathbb{E}[X] = \sum_{\substack{S \subset [n] \\ \#S = k}} \mathbb{P}[K_S \text{ mono.}] = \frac{\binom{n}{k}}{2^{\binom{k}{2}-1}}$$

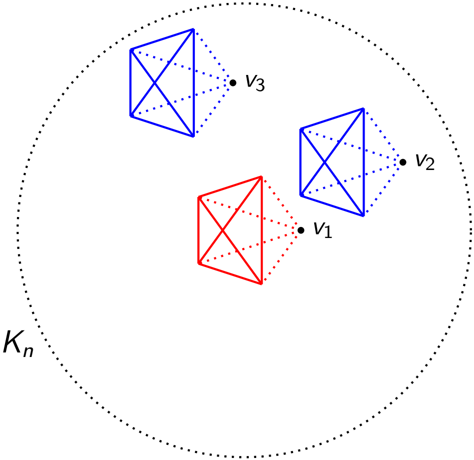
- There exists a coloring with at most $\frac{\binom{n}{k}}{2^{\binom{k}{2}-1}}$ monochromatic K_k.

The alteration method

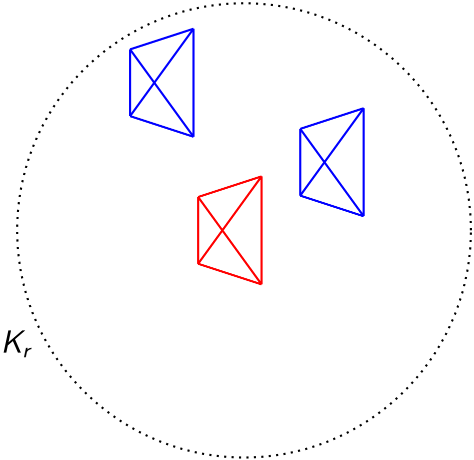


The alteration method

Remove one node from each of the cliques, we obtain a Ramsey coloring on a K_r where $r \geq n - \frac{\binom{n}{k}}{2^{\binom{k}{2}-1}}$



The alteration method



A first lower bound

Corollary

$R(k,k) > \frac{k2^{k/2}}{e}(1+o(1))$

Proof

- Let $n \sim \frac{1}{e}k2^{k/2}(1-o(1))$ to conclude using the previous theorem .

$\square$

A careful analysis shows this simple alteration method doesn't reach any further.

The current best lower bound is given by $R(k,k) > k2^{k/2}(\sqrt{2}/e + o(1))$, which is proved using Lovász local lemma.

Incidence geometry

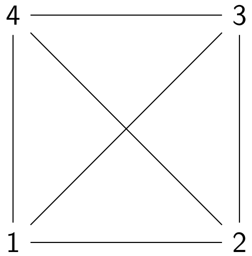
Incidence geometry

Definition

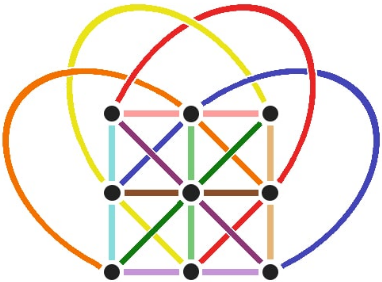
A triplet of sets $(\mathcal{P}, \mathcal{L}, \mathcal{I})$, where $\mathcal{I} \subset \mathcal{P} \times \mathcal{L}$ such that for any $l \in \mathcal{L}$ there are at least two distinct $x, y \in \mathcal{P}$ with $(x, l), (y, l) \in \mathcal{I}$.

The elements of $\mathcal{P}$ are called point, the elements of $\mathcal{L}$ lines and $\mathcal{I}$ is the incidence relation.

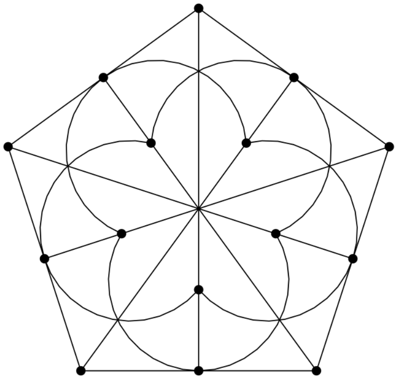
Classical Affine plane of order 2



Classical Affine plane of order 3



Doily



Projective planes

Definition

An incidence geometry, $\pi = (\mathcal{P}, \mathcal{L}, \mathcal{I})$, is said to be a projective plane the following hold:

1. There is a unique line crossing through any two distinct points
2. Any two distinct lines $l, l' \in \mathcal{L}$ intersect at a unique point
3. There exists 4 points no three of which are collinear

Classical projective planes

Definition

The classical projective plane of order q is the incidence geometry, $PG(2,q) = (\mathcal{P}, \mathcal{L}, \mathcal{I})$, whose points are 1 dimensional subspaces and lines the 2 dimensional subspaces of a 3 dimensional $\mathbb{F}_q$ vector space V.

The classical projective plane is a projective plane!

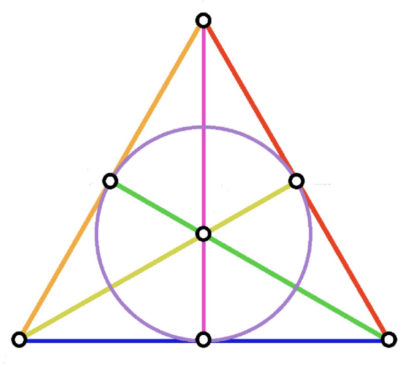
The order of a projective plane

Theorem

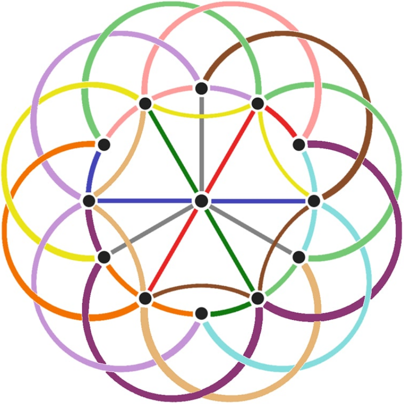
Let $\pi = (\mathcal{P}, \mathcal{L}, \mathcal{I})$ be a projective plane, then there exists $n \in \mathbb{N}_{\geq 2}$ called the order of π such that the following are true:

1. $\#\mathcal{P} = \#\mathcal{L} = n^2 + n + 1$
2. *Through each point crosses $n + 1$ lines*
3. *Each line contains $n + 1$ points*

Fano Plane



Classical Projective plane of order 3



Back to Ramsey Numbers

Equivalent definition of simple Ramsey numbers

$R(s,t) \in \mathbb{N}$ is the minimal integer such that for any graph G on $m \geq R(s,t)$ vertices, G either contains an independent set of size s or a clique of size t.

We shall denote $\alpha(G)$ the size of the largest independent set in G

Triangle free graphs with small $\alpha(G)$

We shall employ Projective planes to give lower bounds on $R(3,t)$.

Theorem

$R(3,t) = \Omega(t^{3/2})$.

Proof:

- Let q be a prime power and $G = (V,E)$ be the incidence graph of $PG(2,q)$.
- G has $2(q^2+q+1)$ nodes and $(q+1)(q^2+q+1)$ edges.
- Consider $\prec$ a linear ordering of E and the graph $H_\prec$ whose vertices are the edges of G and two vertices $pl \prec p'l'$ with $p \neq p', l \neq l'$ are joint by an edge if and only if $pl' \in E$.

$H_\prec$ is triangle free

We must first prove $H_\prec$ is triangle free:

- Suppose three $p_1 l_1 \prec p_2 l_2 \prec p_3 l_3$ all joint by an edge
- Then $p_1 l_2, p_1 l_3, p_2 l_3 \in E$ a contradiction as p_1, p_2 lie both on two distinct lines.

$\alpha(G) < 2(q^2 + q + 1)$

Now let us prove that $\alpha(G)$ is at most quadratic in the order of the plane.

- Let $S \subset E$ be a subset of the vertices of $H_{\prec}$ of size $2(q^2 + q + 1)$.
- As $\#V = 2(q^2 + q + 1)$, there must exist $(p_1, l_1, \ldots, p_k, l_k)$ a cycle in G of size $2k \geq 4$.
- There must exist $i \in \{1, \ldots, k\}$ such that $p_{i+1} l_{i+1} \prec p_i l_i$
- Hence $p_i l_{i+1} \in E$, we deduce $p_i l_i$ and $p_{i+1} l_{i+1}$ must be joint by an edge in $H_{\prec}$.

$\square$

Concluding Remarks

Questions?